

Global Superstore

Executive Analysis



Data Analytics Internship– Final Project

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Domain:Data Analytics

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Task 2: Exploratory Data Analysis (EDA)

In this phase, we perform Exploratory Data Analysis (EDA) to understand the data structure, detect patterns, and identify potential issues before building the final dashboard.

Meaningful Questions

1. **Sales Performance:** What is the overall sales trend across different months, and which months show the highest peak?
 2. **Profitability Analysis:** Which product categories and sub-categories are generating the maximum profit for the store?
 3. **Discount Impact:** How does the level of discount affect the overall net profit? Does high discount lead to negative profit?
 4. **Regional Insights:** Which geographical regions are the top performing market in terms of total sales volume?
 5. **Customer Segmentation:** Which customer segment (Consumer, Corporate, or Home Office) contributes most to the revenue?
 6. **Shipping Efficiency:** What is the average shipping cost, and does it significantly vary across different regions?
 7. **Order Volume:** What is the total count of unique orders processed, and how is it distributed across categories?
 8. **Data Quality Check:** Are there any anomalies, missing values, or negative profit outliers that need to be addressed?
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Key Findings & EDA Insights



After a thorough analysis of the Global Superstore dataset, the following insights were discovered.

1. **Seasonal Peak:** Sales consistently peak during the **fourth quarter (October-December)** due to holiday shopping trends.
2. **Top Category: Technology** is the most profitable category, while **Furniture** shows high sales but lower profit margins due to shipping costs.
3. **Discount Risk:** High discounts (above 20%) are directly linked to **Negative Profit**. We identified several loss-making transactions caused by over-discounting.
4. **Market Leadership: North America and APAC** are the dominant markets, contributing over 60% of total global revenue.
5. **Consumer Power:** The **Consumer Segment** is the largest customer base, followed by the Corporate segment.
6. **Shipping Impact:** Average shipping costs are stable, but international "Standard Class" shipping is the most used method.
7. **Order Volume:** The store successfully processed over **25,000 unique orders**, showing a strong global reach.
8. **Data Quality:** We detected and resolved anomalies in the "Profit" column where extreme negative values were skewing the average.

Phase 2: Data Exploration & Pattern Recognition

1.Data Structure & Variables:

Dataset Overview: The Global Superstore dataset consists of over **51,000 rows** and **24 columns**, providing a granular view of global retail operations.

Data Types: Categorical: Market, Region, Category, and Sub-Category.

- **Numerical:** Sales, Profit, Quantity, and Discount.(used for performance metrics).
- **Temporal:** Order Date and Ship Date(used for time-series analysis).

2.Trends and Patterns (Point 3)

- **Sales Seasonality:** A clear upward trend is identified during the **Q4 (October-December)** period, indicating high seasonal demand.
- **Geographic Concentration:** High sales density is observed in **North America and APAC**, while certain European markets show high profitability despite lower volumes.
- **Anomalies Detected:** We identified “Profit Anomalies” where high sales did not translate to high specific remote regions.

Anomalies & Outliers

- **Anomalies & Outliers:** We identified extreme “Profit Dips” in specific months where high sales were offset by abnormal shipping costs or high return rates.
- **Segment Concentration:** A strong patterns was found where the ‘Consumer’ segment remains the most consistent buyer across all regions.

Hypothesis Testing & Data Challenges

Phase 3: Validation & Data Integrity

Hypothesis Testing & Data validation(Point 4)

✓	Hypothesis “Aggressive discounting (above 20%) is necessary to drive sales volume but has a negative impact on net profit margins.”
✓	Observation By analyzing the Scatter Plot , we confirmed that as discounts increase towards 50%-80%, the profit trend dips sharply into the negative zone.
✓	Validation The assumption is Validated . While high discounts increase the ‘Sum of Quantity’, they are the primary cause of ‘Negative Profit’ across various sub-categories like Tables and Machines.

Detecting Potential Issues (Point 5):

	Negative Profit Outliers: We identified extreme loss-making orders. These are not just business losses but potential "Data Outliers" that could skew the average performance of a region if not analyzed separately.
	Shipping Inconsistencies: Some international orders showed unusually high shipping costs relative to their sales value, indicating a need for more precise "Shipping Mode" data.
	Return Rate Blind spot: The current dataset does not fully account for "Returned Orders," which is a critical problem for further analysis to understand the true Net Profit.
	Standardization: Standardizing the date and currency formats was essential to ensure the KPI cards (Sales & Profit) displayed accurate global figures.

Task 3&4:

The Dashboard Showcase



- **Dynamic Filters:**

User can interactively filter data by **Region** and **Category** to view specific market performance.

- **KPI Monitoring**

Direct tracking of total Sales (12.64M), Profit (1.47M), and Order Volume (25.7k).

- **Profit Visibility:**

Scatter plots and pie charts to understand profitability across segment and discount levels.

Conclusion & Strategic Insights

- **Project Summary:**

The Global Superstore analysis successfully raw data into actionable business intelligence. By utilizing Power BI, we identified critical growth drivers and operational risks.

- **Key Takeaway:**

The most significant finding was the direct correlation between high discounts and negative profit. Addressing this through a “Tiered Discounting Strategy” can immediately improve the store’s bottom line

- **Final Recommendations**

Prioritize expansion in high profit regions like North America and APAC.

Cost Control: Review shipping logistics for heavy volume categories (Furniture) to minimize overheads.

Retention: Implement loyalty programs for the ‘Consumer’ segment to maintain steady revenue growth.

- **Final Statement:**

This internship project provided hands on experience in data cleaning, EDA, and advanced visualization, demonstrating the vital role of data analytics in modern decision making.

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